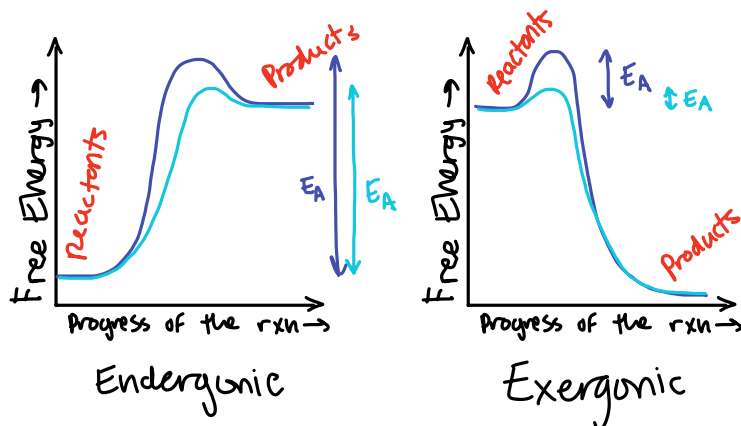


unit 3

- **Metabolism**—sum of organism's chemical reactions
- **Metabolic pathway**—series of reactions that takes an initial substance and makes a final product, using enzymes
- **Specificity of Enzymes**—only one specific tertiary structure will fit into the active site of an enzyme
- **Catabolic pathway**—net release of energy, breakdown of molecules
- **Anabolic pathway**—net consumption of energy, synthesis of molecules
- **Kinetic energy**—energy associated with the movement of matter
- **Potential energy**—energy matter possesses because of location or structure; what the matter can do but is not currently doing
- **Thermodynamics**—the study of energy transformations
- **First law of thermodynamics**—energy cannot be created nor destroyed, only transformed/transferred
- **Second law of thermodynamics**—every energy transformation/transfer increases the entropy of the universe
- **Entropy**—measure of disorder or randomness in the universe
 - *** Living systems increase the entropy of the universe around them more than they decrease the entropy of the universe around them
- **Free energy (G)**—portion of a system's energy that is readily available to do work
- **Change in Free Energy (ΔG)**— $G_{\text{final}} - G_{\text{initial}}$ (how much free energy in the system before and after the reaction)
- **Spontaneous Process**—a reaction happens randomly because $\Delta G < 0$, exergonic reaction, decreases free energy in the system, stability of the system increases
 - *** The amount of free energy in a system and the stability of the system are inversely related
- **Exergonic Reaction**—releases energy, $\Delta G < 0$ (negative), products have less free energy than reactants, happens spontaneously, catabolic
- **Endergonic Reaction**—stores energy, $\Delta G > 0$ (positive), products have more free energy than reactants, cannot happen spontaneously because free energy is required from the environment, anabolic
- **Activation Energy (E_A)**—the initial amount of energy reactants need to absorb to become products. Enzymes reduce the amount of activation energy required for a reaction.



- Without Enzyme
- With Enzyme

*** All organisms require a system in which energy input exceeds energy loss. Otherwise, they will die.

- **Reaction Coupling**—an exergonic reaction releases energy that is used in an endergonic reaction. The absolute value of the exergonic ΔG must be larger than the value of the endergonic ΔG . Net reactions must be exergonic, total $\Delta G < 0$.
- **ATP (Adenosine Triphosphate)**—a molecule that stores lots of energy between the second and third phosphate
- **ATP hydrolysis**—ATP is split into ADP, an independent phosphate (P_i), and releases free energy
- **ATP coupling**—an endergonic reaction is coupled to ATP hydrolysis
- **ADP phosphorylation**—ADP and an independent phosphate (P_i) combine in an anabolic reaction to make ATP
- **Endotherm**—maintains a stable internal body temperature using metabolic reaction
- **Ectotherm**—relies on its environment's temperature to determine its metabolic rate
- **Redox Reactions (LEO goes GER)**—oxidation and reduction reactions that occur when electrons are transferred (typically with hydrogen atoms)
- **Oxidation (LEO-Loses Electrons, Oxidation)**—proportion of oxygen increases, proportion of hydrogen decreases, loses electrons
 - *** In cellular respiration, glucose is oxidized into CO_2 .
- **Reduction (GER-Gains Electrons, Reduction)**—proportion of oxygen decreases, proportion of hydrogen increases, gains electrons
 - *** In photosynthesis, oxygen is reduced to H_2O .
- **Electron Carrier Molecules**—molecules like FAD and NAD^+ can carry electrons with hydrogen atoms from one part of a system (cellular respiration/photosynthesis) to another
 - *** $FAD + 2e^- \Rightarrow FADH_2$
 - *** $NAD^+ + 2e^- \Rightarrow NADH$
- **Coenzyme**—non-protein helper of an enzyme
- **Proton-motive force**—protons want to move across a membrane when imbalanced, moving from high to low concentration
- **Substrate-Level Phosphorylation**—a substrate transfers a high-energy phosphate directly to ADP (used during glycolysis)
- **Aerobic Respiration**—uses oxygen to fully break down glucose, oxygen will be the last electron acceptor
 - **Glycolysis**—splits one 6-carbon molecule of glucose into two intermediate 3-carbon G3P molecules and then into two 3-carbon pyruvate. Occurs in the cytoplasm outside of the mitochondria.
 - **Inputs:** Glucose, 2ATP, 4ADP, 2 NAD^+
 - **Outputs:** 2 Pyruvate, NET 2 ATP, 2 NADH
 - **Pyruvate Oxidation**—Coenzyme A (CoA) is added to the two pyruvate molecules in the mitochondrial matrix, producing 2 Acetyl-CoA molecules, 2 Carbon Dioxide, and 2 NADH.
 - **Citric Acid/Krebs Cycle**—Acetyl-CoA binds to oxaloacetate to form citrate until citrate eventually oxidizes back into oxaloacetate. Occurs in the mitochondrial matrix, twice, because there are two Acetyl-CoA molecules.
 - **Inputs:** 2ADP, 6 NAD^+ , 2FAD, 6 H_2O , 2 Acetyl-CoA
 - **Outputs:** 2ATP, 6NADH, 2 $FADH_2$, 4 CO_2

► **Oxidative Phosphorylation**—uses FADH_2 and NADH carriers to produce lots more ATP.

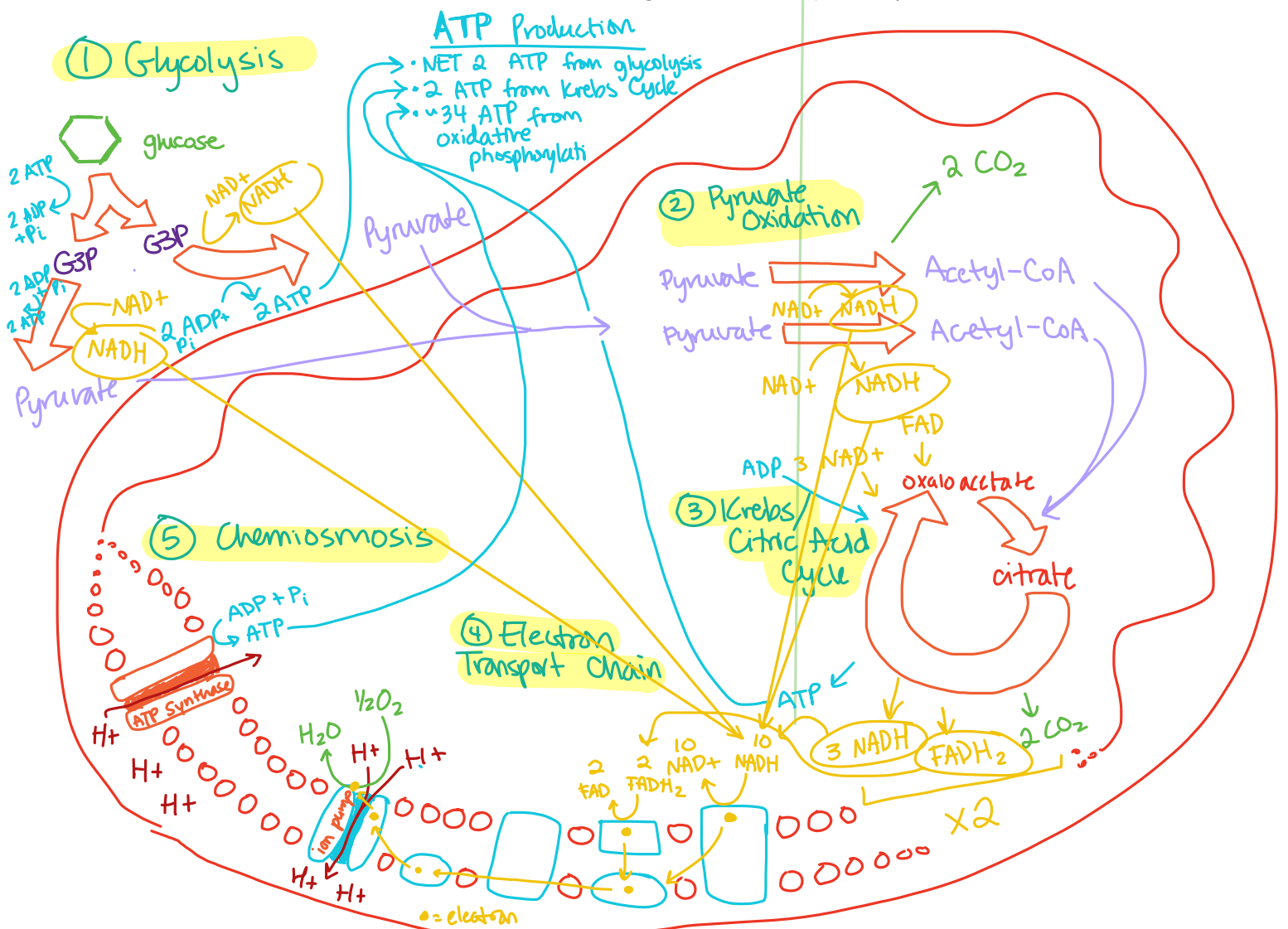
- **Electron Transport Chain**—Occurs on the inner mitochondrial membrane. NADH AND FADH_2 drop off their electrons at protein complexes, which then moves the high-energy electrons to ion pumps, which pump H^+ ions against the concentration gradient to the intermembrane space (active transport). Oxygen atoms then accept the electrons to become H_2O .

- Each FADH_2 pumps out 2 H^+ ions, whereas NADH pumps out 3 H^+ ions.

- **Chemiosmosis**—a proton-motive force pushes H^+ ions through a channel protein called ATP Synthase, which allows the H^+ ions to pass to low concentration while at the same time generating an ATP molecules from an ADP and a P_i with every diffusion of an H^+ ion.

- **Oxidative Phosphorylation produces roughly 34 ATP.** ($10 \text{ NADH} \times 3 + 2 \text{ FADH}_2 \times 2$)

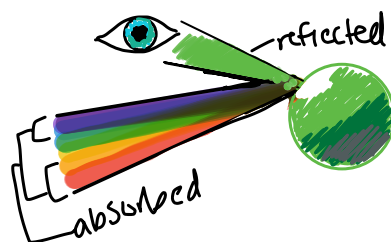
- **Decoupling Oxidative Phosphorylation**—other channel proteins can allow H^+ ions to passively diffuse, merely generating heat instead of producing ATP. This allows endothermic to regulate body temperature.



*** Aerobic cellular respiration is performed by both aerobic prokaryotes and most eukaryotes (in the mitochondria).

*** Additionally, glycolysis can accept other molecules such as carbohydrates, fat, or even proteins.

- **Feedback inhibition**—when ATP is present, it can act as an inhibitor to glycolysis or other parts of cellular respiration to ensure that there is no overproduction of ATP.
- **Glycolysis as Evolutionary Proof**—glycolysis is the most common metabolic pathway on earth and offers proof that all organisms who can perform glycolysis share an ancestor who once developed the glycolysis trait.
- **Anaerobic respiration**—another molecule (sulfur, nitrogen...) is used as the final electron acceptor instead of oxygen in cellular respiration
- **Fermentation**—coupled with glycolysis to cycle NAD^+ to produce a NET 2 ATP from every glucose molecule (much more inefficient than aerobic respiration).
 - **Alcoholic Fermentation**—pyruvate is converted to ethanol while turning 2 NADH into 2 NAD^+ (used in yeast cells). Releases 2 CO_2 .
 - **Lactic Acid Fermentation**—pyruvate is converted to lactate without releasing CO_2 while regenerating NAD^+ (used in human muscles when oxygen is scarce).
- **Anaerobe**—organism that can survive on anaerobic pathways
- **Facultative anaerobe**—can toggle between aerobic and anaerobic pathways
- **Obligate anaerobe**—cannot survive in the presence of oxygen
- **Autotroph**—organism that produces its own food source
- **Chemoautotroph**—builds organic compounds from inorganic chemical compounds
- **Photoautotroph**—builds organic compounds using light energy
- **Heterotroph**—consumes other organisms for energy
- **Stomata**—pores on leaves where CO_2 enters and O_2 leaves, as controlled by the opening and closing of guard cells.
- **Chloroplast**—organelle where plants perform photosynthesis
- **Mesophyll**—interior tissue of the leaf where cells contain more chloroplasts.
- **Thylakoids**—connected, membrane-bound sacs of chlorophyll in chloroplasts
- **Grana (singular=granum)**—columns of thylakoids
- **Stroma**—dense fluid found outside of thylakoids in the inner membrane.
- **Photon**—discrete energy particle that makes up light
- **Pigment**—substance that absorbs certain wavelengths of visible light



- **Photosynthesis**—the process by which plants and other photoautotrophs build chemical compounds (glucose) by harnessing light energy
 - **Light-Dependent Reactions**—the stage of photosynthesis that is dependent on the absorption of light by photosystems to produce ATP and NADPH

- **Photosystem II**—light energy captured by chlorophyll pigments excites electrons, an electron acceptor will carry the high-energy electrons through the electron transport chain. The light energy captured by photosystem II will be used to split a water molecule into two H^+ ions, an oxygen atom, and two electrons.
- **Electron Transport Chain (ETC)**—electron energy is used to pump H^+ ions into the thylakoid lumen and out of the stroma. High H^+ concentration inside the thylakoid lumen.
- **Photosystem I**—electrons are reenergized by light energy captured by photosystem I and are sent down to NADP+ Reductase.
- **NADP+ Reductase**—electrons are then given to the final electron carrier of the ETC, NADP+, which becomes NADPH, and will carry the electrons to the light-independent reactions.
- **Chemiosmosis**— H^+ ions passively diffuse to the stroma through ATP synthase, which joins ADP and a P_i to form ATP with each ion that passes through.
- ▶ **Light-Independent Reactions (Calvin Cycle)**—no light is required, but the products from the light-dependent reactions are required. CO_2 is turned into 3-PGA, which then becomes G3P, which can then become glucose.
 - **Carbon Fixation**—three carbon dioxide molecules are fixed to three RuBP to form 6 3-PGA molecules.
 - **Reduction**—NADPH reduces the 3-PGA molecules into 6 G3P molecules. 5 will stay until regeneration, and one will be used for production of glucose, lipids, or amino acids. Two G3P are needed to form one glucose molecule.
 - **RuBP regeneration**—the remaining 5 G3P molecules are turned back into three RuBP, which requires ATP (from ATP synthase).

